

## SECTION 4.2

- Electronics: The main components of microcontrollers are, communication modules, sensors, real-time clocks (RTC), and storage devices. The most common controller is Arduino, explicitly used in 29 papers, followed by Raspberry Pi in 10 and NodeMCU in 2. Arduino is popular because it is flexible, low-cost, and available in models such as Mega, Uno, and Nano. Raspberry Pi offers greater processing capability and can connect motors and sensors through GPIO while also supporting medication schedules, event monitoring, alerts, displays, keypads, and data storage. NodeMCU is an IoT-oriented ESP8266 board with integrated Wi-Fi and can send notifications, monitor medication levels, issue refill alerts, and synchronize data with mobile apps or web servers.
- Communication technologies: The main options are GSM, Wi-Fi, Bluetooth, and NFC. GSM can send SMS messages or calls to patients/caregivers, including alerts for missed doses or low medication levels. Wi-Fi enables remote access and monitoring through online services. Bluetooth is mainly used to connect the dispenser with a smartphone/tablet, transfer medication schedules, change settings, and exchange therapy information. NFC appears less frequently and is used, for example, to verify a medicine package.
- Sensors, timing, and storage: IR sensors can detect the presence of a user's hand or whether a pill is present. Cameras are used for pill recognition, prescription/OCR processing, monitoring medication-taking activity, or communication. Kinect-based systems use cameras and depth sensors to track the patient's movement while taking medication, while fingerprint sensors provide authentication. The RTC is important for maintaining accurate time, triggering medication events at scheduled times, and recording the previous dispensing time to help prevent over- or under-dosing. SD cards are used to save medication/dispensing data for later retrieval.
- Mechanics: Mechanical elements provide the actual movement needed for automatic dispensing. The paper discusses stepper motors, servo motors, brush

motors, and motor drivers. Stepper motors are particularly useful because they move in precise predefined steps, allowing a dispenser to release a controlled quantity of medication. Servo motors provide precise position control, while the motor driver controls the motor movement.

- Software: Software is divided into application level and system/infrastructure level. Mobile and web applications are widely used to manage medication schedules, send reminders, monitor device status, and synchronize information with caregivers or doctors; Android/web applications appeared in 22 papers. Platforms such as Blynk allow remote control of IoT devices and customized interfaces, while ThingSpeak can collect and visualize sensor/device data. Cloud services store medication data and make it remotely accessible. At system level, MQTT provides lightweight communication between the dispenser, mobile apps, and cloud services, while ROS is used in robotic dispenser systems to coordinate sensors and actuators.

#### SECTION 4.4

- Direct interaction with the device: The most common physical interfaces are buttons/keypads, LEDs, LCDs, speakers, buzzers, and touchscreens. Buttons and keypads are mainly used to control the device or set medication times. LEDs are widely used for simple status indications, while LCDs show information such as medication name, time, remaining pills, or dosage. Speakers and buzzers provide audible reminders or alerts. Touchscreens combine input and output in one interface.
- Authentication and accessibility: Some systems use fingerprint recognition or facial recognition to ensure that only authorized users can access the medication. A few systems also use microphones and voice interaction, especially for visually impaired users, although the paper notes that such voice interaction is still limited and often not very natural.
- Interaction through separate devices: Many solutions use an external device, especially a smartphone, to interact with the dispenser. Smartphones can be used to set medication schedules, receive notifications, monitor medication status, control the device, and sometimes manage refilling. Personal computers, mobile robots, and wearable devices are also used, but much less frequently.
- Mobile applications and notifications: Mobile/web apps are commonly used for setting medication times, receiving reminders, monitoring adherence, and

notifying caregivers or healthcare professionals when a dose is missed. Some systems use SMS instead of a dedicated app.